

Perceptual Data Structures

A Cognition-Inspired Event Processing and Emotional Feedback System

Abstract

This report presents the design and implementation of a cognition-inspired system called *Perceptual Data Structures*. The system models how an artificial agent can perceive events, associate them with emotion-based structures, and refine its internal decisions through user emotional feedback. It combines semantic similarity, emotional similarity, memory formation, and adaptive thresholding to mimic mechanisms inspired by human perception and learning. The report explains all system components in accessible, non-technical language while remaining faithful to the underlying computational ideas.

1 Introduction

Humans perceive the world through a blend of reflexive reactions and slower, context-driven reasoning. The project *Perceptual Data Structures* attempts to simulate this behaviour in a computational system. It allows the model to receive events as input, interpret them using both semantic meaning and emotional state, store them in memory, and continuously adapt based on user-provided emotional signals.

This project integrates:

- a perception model structured around fast and slow decision pathways,
- a circumplex emotion interface for user feedback,
- a cloud database for memory persistence, and
- an interactive Streamlit dashboard for model visualisation.

The goal is not to mimic human cognition precisely, but to explore whether perceptual structures (such as emotional similarity and semantic proximity) improve machine interpretation and adaptation.

2 Problem Statement

Conventional event-processing systems rely heavily on fixed rules or machine-learned weights. They often ignore emotional context, semantic similarity, or incremental user feedback. This limits how effectively they adapt to nuanced inputs.

The problem addressed in this project is:

How can an artificial system build and update perceptual structures by combining semantic meaning, emotional context, and user feedback?

The system must:

- handle incoming events with labels, intensity, and context,
- identify whether new events resemble previously-seen ones,
- assign emotional meaning using a valence–arousal model,
- store experiences in memory and update thresholds,
- integrate user emotional corrections, and
- maintain an interpretable trail of decisions.

3 System Architecture

The architecture consists of the following major components:

1. Brain-Inspired Model

The central cognitive engine that:

- processes incoming events,
- compares them against memory,
- computes semantic and emotional similarity,
- updates internal thresholds,
- stores experiences for future reference.

2. Flask Backend

A lightweight API layer exposing:

- endpoints for event processing,
- endpoints for user emotional feedback,
- UI pages for interaction.

3. MongoDB Atlas Database

A cloud-hosted storage layer that logs:

- every event processed,
- every user emotional submission,
- memory history and differences between model and user.

4. Circumplex Emotion UI

A front-end wheel allowing users to express emotional responses through:

- angle selection (emotion),
- valence (positive–negative),
- arousal (calm–excited),
- certainty (via hold time).

5. Streamlit Dashboard

Provides interactive visualisation of:

- thresholds,
- valence–arousal distributions,
- confidence differences,
- recent events and memory evolution.

4 Core Model Logic

The cognitive model uses two complementary pathways similar to human decision systems.

4.1 Fast Path (Reflexive Decision)

This is a quick, threshold-based response. If the incoming event intensity exceeds the current threshold, the model selects a “RUN” response; otherwise it chooses “STAY.” This part is intentionally simple because reflexive reactions in biological systems are often immediate.

4.2 Slow Path (Semantic and Emotional Interpretation)

When processing an event more thoughtfully, the system evaluates:

- its semantic meaning, by comparing textual labels,
- its emotional similarity to past experiences.

The model computes two kinds of similarity:

1. Semantic Similarity

Events with similar wording or meaning should be treated as related. The system uses high-level sentence embeddings to measure how close two labels are in meaning.

2. Emotional Similarity

If two events typically produce similar valence (positive/negative feeling) or arousal (intensity), the model considers them emotionally related.

These two forms of similarity are blended into a combined score. If a new event is highly similar to past ones, the model “borrows” emotional expectations from those earlier experiences. This simulates how humans associate new situations with old memories.

4.3 Memory Structure

For each distinct event label, the model stores:

- how often it was safe or dangerous,
- historical valence values,
- historical arousal values.

This allows the model to develop an evolving emotional map of all events it has perceived.

4.4 Adaptive Threshold

The model’s threshold dynamically changes based on outcomes. For example:

- If many events labelled “danger” occur, the threshold decreases, making the agent more cautious.
- If many “safe” events occur, the threshold rises, indicating increasing trust.

This provides a simple but effective feedback mechanism.

5 Event Processing Pipeline

The full event pipeline proceeds as follows:

1. A user enters an event label, intensity, and context.
2. The fast path immediately evaluates the reflexive response.
3. The slow path checks semantic and emotional similarity.
4. The model consolidates both results to produce a final response.
5. Memory and thresholds are updated accordingly.
6. The entire event is stored in the database with time-stamping.

This ensures that each event contributes to the model's ongoing learning process.

6 Circumplex Emotion Integration

The circumplex model is a well-established representation of human emotions in two dimensions:

- **Valence:** positive versus negative feeling,
- **Arousal:** calm versus excited.

Users select a point on a circular wheel. This determines:

- an angle (representing an emotion category),
- a valence value,
- an arousal value,
- a certainty level (based on hold time).

These values form the basis of how the system learns from human emotional interpretation. This also forms the basis of a feedback loop which ensures continuous refinement.

7 Collision Logic

The collision mechanism determines whether a new event resembles an existing memory.

A collision happens when the combined semantic and emotional similarity to past events exceeds a threshold. This indicates that the event shares meaning or emotional signature with past experiences.

If a collision is detected:

- the model acknowledges the new event as non-novel,

- emotional values from previous events are borrowed,
- memory grows richer and more interconnected over time.

This makes the system behave more like a perceptual organism, forming mental clusters of related experiences.

8 Database Design

A cloud-hosted MongoDB Atlas database stores all perceptual logs. Each entry contains:

- event metadata,
- model decisions,
- emotional values,
- threshold values,
- user feedback,
- historical emotion arrays.

The database effectively becomes the system's long-term perceptual memory.

9 Flask Backend

The backend performs the following key roles:

- serves the circumplex and event UIs,
- handles emotional submissions,
- pushes data into MongoDB,
- provides model state for dashboard visualisation.

It acts as the bridge connecting UI, model, and database.

10 Frontend Design

Event Interface

Users can enter labels, intensities, and contexts. The system responds with reflexive and final decisions, displayed in real time.

Circumplex Interface

An interactive circle lets users visually express their emotional reactions. This produces intuitive feedback input without requiring technical knowledge.

11 Streamlit Dashboard

The dashboard visualises:

- threshold changes,
- model confidence distributions,
- valence–arousal scatter plots,
- confidence differences between model and user,
- recent events and their emotional trajectories.

It serves as the analytical window into the evolving perceptual state of the model.

12 Data Flow Diagram (Textual)

1. User submits an event through the UI.
2. Backend forwards it to the model.
3. Model processes it using fast and slow reasoning.
4. The resulting perceptual record is stored in the database.
5. User may submit circumplex feedback.
6. Model adjusts its memory and threshold accordingly.
7. Dashboard fetches records for visualisation.

13 Conclusion

The project demonstrates how perception, emotion, memory, and learning can be integrated into a unified computational system. The *Perceptual Data Structures* model adapts continuously as new events and feedback arrive, forming meaningful associations between experiences through semantic and emotional similarity. The combination of backend, database, and visual analytics creates a complete pipeline for studying cognitive behaviour in artificial agents.

14 Future Work

Future enhancements may include:

- adding forgetting mechanisms to prevent over-saturation,
- integrating sensor-based input such as audio or natural text,
- implementing multi-step reinforcement learning.

These improvements would advance the system's realism and make it more suitable for adaptive agents.